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1. Introduction

Weather data is generated continuously from thousands of weather stations around the world. Every day, these stations record different weather conditions such as maximum temperature, minimum temperature, average temperature, rainfall, snowfall, wind speed, and other atmospheric measurements. As the number of observations increases, processing such a large amount of weather data using traditional data-processing techniques becomes difficult and time-consuming.

Big Data technologies provide an efficient way to store and process such large datasets. Hadoop is one of the most widely used frameworks for distributed storage and processing of large-scale data. It provides the Hadoop Distributed File System (HDFS) for storing large datasets and MapReduce for processing them across multiple nodes or tasks.

In this project, Hadoop was used to analyze a large-scale weather dataset containing daily weather observations from thousands of weather stations. The project focuses mainly on temperature and precipitation-related information. Apache Pig was also used to perform similar weather analyses using a high-level data-flow language.

The main purpose of this project is to understand how Hadoop can be applied to a real-world Big Data problem. Different weather patterns were analyzed, including average maximum temperature, total rainfall, monthly rainfall, and the coldest days of the year.

2. Problem Statement

The weather dataset used in this project contains millions of records collected from a large number of weather stations. Each record contains information about weather observations for a particular station and date.

Processing millions of records manually or using conventional data-processing methods can be inefficient. Therefore, the problem addressed in this project is to design a Hadoop-based system that can efficiently process the large weather dataset and extract useful information from it.

The project aims to answer questions such as:

- What is the average maximum temperature?
- How does rainfall vary from month to month?
- Which days were the coldest according to the minimum temperature?
- How can these calculations be performed efficiently using Hadoop MapReduce and Apache Pig?

3. Objectives

The main objectives of this project are:

- To store the large weather dataset using HDFS.
- To process weather data using MapReduce.
- To analyze weather data using Apache Pig.
- To calculate average maximum temperature.
- To calculate total and monthly rainfall.
- To identify the top 10 coldest days.
- To understand the practical use of Hadoop for Big Data analysis.

4. Dataset Description

The dataset used in this project is based on the NOAA Global Historical Climatology Network Daily (GHCN-Daily) weather database. GHCN-Daily is a large collection of daily climate observations obtained from land-based weather stations around the world. The database contains different meteorological measurements such as maximum temperature, minimum temperature, average temperature, precipitation, snowfall, snow depth, wind-related observations, and other weather measurements.

For this project, the selected dataset contains weather observations for the complete year **2020**, covering the period from **January 1, 2020 to December 31, 2020**.

After examining the actual dataset used in this project, it contains:

- **Total records:** 36,837,019
- **Weather stations:** 42,793
- **Time period:** January 1, 2020 – December 31, 2020
- **Data format:** Compressed GZIP dataset
- **Data type:** Daily weather observations

Dataset Source: NOAA National Centers for Environmental Information (NCEI)

Official Dataset Website:

<https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily>

5. Technologies and Tools Used

The following technologies and tools were used in this project:

5.1 Hadoop

Apache Hadoop was used as the main Big Data processing framework. Hadoop provided the environment required for distributed storage and processing of the weather dataset.

5.2 HDFS

Hadoop Distributed File System (HDFS) was used to store the large weather dataset. The dataset was uploaded from the local computer to HDFS before processing.

5.3 MapReduce

MapReduce was used to process the weather records and perform different calculations. The Map phase extracts and organizes relevant values, while the Reduce phase combines the values and produces the final result.

5.4 Apache Pig

Apache Pig was used as a high-level data-flow platform for processing the weather dataset. Pig Latin scripts were written to perform weather calculations without manually implementing every MapReduce operation.

5.5 Windows Command Prompt

Windows Command Prompt was used to execute Hadoop, HDFS, MapReduce, and Pig commands.

6. HDFS Implementation

The large weather dataset was first transferred from the local computer to HDFS. HDFS provides distributed storage for large files and is designed to handle large-scale datasets.

The `jps` command was used to verify the running Hadoop Java processes. The output confirms that the required Hadoop services were successfully started and ready for data processing.

```
C:\Windows\System32>jps
23296 NodeManager
17668 NameNode
18836 DataNode
20764 ResourceManager
7356 JobHistoryServer
8460 Jps
```

After uploading the dataset, the HDFS directory was checked to make sure that the input data was available before starting the analysis. Typical HDFS operations used in the project included:

- Creating an HDFS directory
- Uploading the dataset
- Checking the uploaded file
- Running MapReduce jobs using the HDFS input
- Storing output files in HDFS
- Viewing the generated results

The HDFS input directory was used as the source for the Hadoop processing jobs.

```
C:\hadoop\bin>hdfs dfs -mkdir /weather

C:\hadoop\bin>hdfs dfs -mkdir /weather/input

C:\hadoop\bin>hdfs dfs -mkdir /weather/output

C:\hadoop\bin>hdfs dfs -mkdir /weather/raw

C:\hadoop\bin>hdfs dfs -ls /
Found 6 items
drwxr-xr-x - BS456 supergroup          0 2026-08-16 12:49 /weather_output
drwxr-xr-x - BS456 supergroup          0 2026-08-16 11:56 /bda3
drwxr-xr-x - BS456 supergroup          0 2026-08-16 18:01 /pigdata
drwx----- - BS456 supergroup          0 2026-08-26 19:18 /tmp
drwxr-xr-x - BS456 supergroup          0 2026-09-01 19:24 /weather
drwxr-xr-x - BS456 supergroup          0 2026-08-22 22:05 /youtubepig
```

7. Apache Pig Analysis

Apache Pig was used to perform the same types of weather analysis using Pig Latin. Pig provides a simpler way to express data-processing operations because the developer can describe the required data flow instead of manually implementing every MapReduce operation.

Ten Pig analyses were performed:

1. Average Maximum Temperature
2. Average Minimum Temperature
3. Extremely Hot days
4. Hottest Days
5. Monthly Rainfall
6. Monthly Temperature
7. Monthly Extreme Hot days
8. Top 10 Coldest Days
9. Top 10 Hottest Days
10. Top 10 Rainiest Days

7.1. Average Maximum Temperature

The Average Maximum Temperature analysis was performed using Apache Pig to calculate the average of the maximum temperature values recorded in the weather dataset. The Pig script filters the required temperature records and calculates the average value using Pig's aggregation functions. This analysis helps to understand the overall maximum temperature pattern in the dataset.

Pig script for Average Maximum Temperature:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>pig -x mapreduce pig\avg_max_temperature.pig
2026-09-01 19:53:49,579 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-01 19:53:49,581 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-01 19:53:49,581 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-01 19:53:50,529 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-01 19:53:50,529 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788270830518.log
2026-09-01 19:53:51,829 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\BS456/.pigbootstrap not found
2026-09-01 19:53:51,917 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-01 19:53:51,918 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-01 19:53:52,914 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-avg_max_temperature.pig-2df934a2-f818-4313-9203-a3623f1d4893
2026-09-01 19:53:52,914 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-01 19:53:54,334 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-01 19:53:54,441 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set - will not generate code
```

Output of Average Maximum Temperature:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs -ls /weather/output

Found 1 items
drwxr-xr-x - BS456 supergroup          0 2026-09-01 19:59 /weather/output/avg_max_
temperature

C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs -ls /weather/output
/avg_max_temperature
Found 2 items
-rw-r--r--  3 BS456 supergroup          0 2026-09-01 19:59 /weather/output/avg_max_
temperature/_SUCCESS
-rw-r--r--  3 BS456 supergroup        19 2026-09-01 19:59 /weather/output/avg_max_
temperature/part-r-00000

C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs -cat /weather/outpu
t/avg_max_temperature/part-r-00000
16.929993069965775
```

7.2. Average Minimum Temperature

The Average Minimum Temperature analysis was performed using Apache Pig to calculate the average minimum temperature from the weather records. The analysis provides an overall view of the minimum temperature conditions recorded during the study period.

Pig script for Average Minimum Temperature:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>pig -x mapreduce pig\avg_min
_temperature.pig
2026-09-01 20:04:03,782 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-01 20:04:03,801 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-01 20:04:03,801 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-01 20:04:04,745 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0
(r1797386) compiled Jun 02 2017, 15:41:58
2026-09-01 20:04:04,745 [main] INFO org.apache.pig.Main - Logging error messages to
: C:\hadoop\logs\pig_1788271444736.log
2026-09-01 20:04:06,231 [main] INFO org.apache.pig.impl.util.Utils - Default bootup
file C:\Users\BS456/.pigbootup not found
2026-09-01 20:04:06,317 [main] INFO org.apache.hadoop.conf.Configuration.deprecatio
n - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-01 20:04:06,319 [main] INFO org.apache.pig.backend.hadoop.executionengine.H
ExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-01 20:04:07,177 [main] INFO org.apache.pig.PigServer - Pig Script ID for th
e session: PIG-avg_min_temperature.pig-de3c0c8c-594c-475c-a33f-64f34fc80860
```

Output for Average Minimum Temperature:

output/avg_min_temperature

Block ID: 1073742211
Block Pool ID: BP-727263249-192.168.246.1-1786856512627
Generation Stamp: 1387
Size: 19
Availability:

- Mahir

File contents

5.5088932742390675

7.3. Extremely Hot Days

The Extremely Hot Days analysis was performed using Apache Pig to identify days where the maximum temperature exceeded the selected threshold. This analysis helps to identify periods with unusually high temperatures in the dataset.

Output for Extremely Hot Days:

/output/ExtremeHotDays

Block ID: 1073742866
Block Pool ID: BP-727263249-192.168.246.1-1786856512627
Generation Stamp: 2042
Size: 4957212
Availability:

- Mahir

File contents

ASN00016098,20200101,36.4
ASN00017043,20200101,37.9
ASN00017110,20200101,36.0
ASN00017123,20200101,38.3
ASN00017126,20200101,37.6
ASN00053115,20200101,39.5
ASN00054038,20200101,41.1
ASN00055049,20200101,42.0
ASN00055136,20200101,36.0

The following screenshot shows the output containing the days identified as extremely hot according to the defined temperature condition.

7.4. Hottest Days

The Hottest Days analysis was performed using Apache Pig to identify the days with the highest recorded maximum temperatures. The result provides a clear view of the hottest weather conditions found in the dataset.

Output for Hottest Days:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs
-cat /weather/output/hottest_days/part-r-00000
USR0000CCLV,20200828,65.6
USR00000ONI,20200129,64.4
USR0000CCLV,20200904,62.8
USR00000ONI,20200514,62.2
USR00000ONI,20200513,61.7
USR0000CCLV,20200830,60.0
ASN00014627,20200429,57.5
JSC00026482,20200805,56.1
JSC00042319,20200816,54.4
JSC00042319,20200712,53.3
```

The screenshot shows the Pig output showing the days with the highest recorded maximum temperatures.

7.5. Monthly Rainfall

The Monthly Rainfall analysis was performed using Apache Pig to calculate the total rainfall for each month. The date information was used to group the rainfall records by month, allowing the rainfall pattern throughout the year to be observed.

Output of Monthly Rainfall:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs
-cat /weather/output/monthly_rainfall/part-r-00000
01,2466055.800000584
02,2574660.699999527
03,2447780.9000023957
04,2312457.2000010903
05,2904229.1999972006
06,2612985.4000012456
07,2925610.8999991682
08,2660050.799999441
09,2585913.7000005282
10,2413988.0000003753
11,2036396.2000007962
12,2403635.0000013397
```

The following screenshot shows the monthly rainfall results generated by Apache Pig after grouping the rainfall records by month.

7.6. Monthly Temperature

The Monthly Temperature analysis was performed using Apache Pig to analyze temperature values month by month. The records were grouped according to the month so that the variation of temperature throughout the year could be observed.

Output of Monthly Temperature:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs -cat
/weather/output/monthly_temperature/part-r-00000
01,6.701278072746618
02,8.083152915922096
03,11.855725139023587
04,15.75822016773871
05,20.502360383002703
06,24.956610488367282
07,27.391136685421465
08,27.32859064253722
09,23.26237193062776
10,17.201821634380696
11,12.594025086643065
12,7.3390727100785424
```

The screenshot below presents the monthly temperature results obtained from the weather dataset using Apache Pig.

7.7. Monthly Extreme Hot Days

The Monthly Extreme Hot Days analysis was performed using Apache Pig to determine the number of extremely hot days in each month. This analysis helps to identify which months experienced a higher number of extreme temperature events.

Output of Monthly Extreme Hot Days:

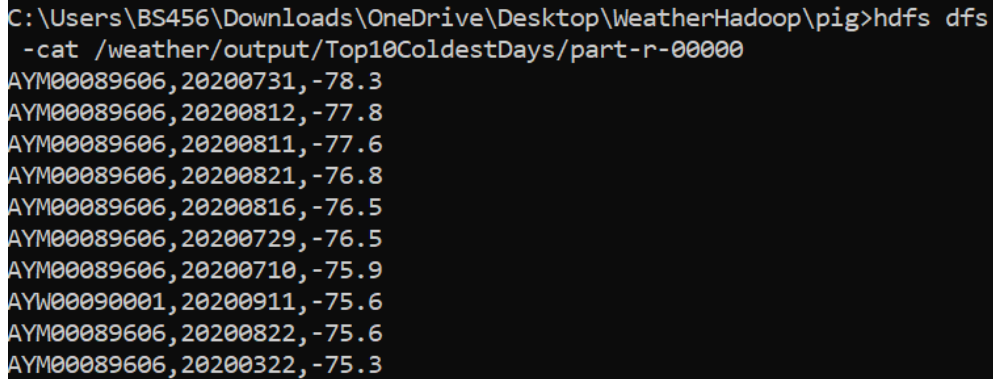
```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop\pig>hdf
s dfs -cat /weather/output/MonthlyExtremeHot/part-r-00000
01,7624
02,5691
03,6078
04,7342
05,12606
06,23215
07,44058
08,44986
09,17230
10,9833
11,6758
12,5241
```

The following screenshot shows the number of extremely hot days identified for each month using Apache Pig.

7.8. Top 10 Coldest Days

The Top 10 Coldest Days analysis was performed using Apache Pig by sorting the temperature records in ascending order and selecting the ten lowest temperature values. This provides the ten coldest recorded days in the dataset.

Output of Top 10 Coldest Days:



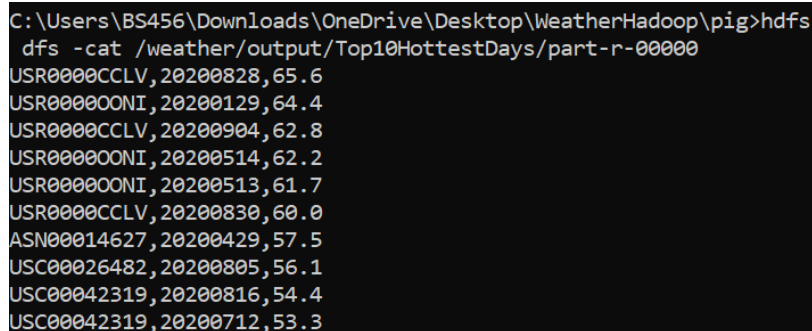
```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop\pig>hdfs dfs
-cat /weather/output/Top10ColdestDays/part-r-00000
AYM00089606,20200731,-78.3
AYM00089606,20200812,-77.8
AYM00089606,20200811,-77.6
AYM00089606,20200821,-76.8
AYM00089606,20200816,-76.5
AYM00089606,20200729,-76.5
AYM00089606,20200710,-75.9
AYW00090001,20200911,-75.6
AYM00089606,20200822,-75.6
AYM00089606,20200322,-75.3
```

The screenshot below presents the Pig output showing the 10 coldest days identified from the weather dataset.

7.9. Top 10 Hottest Days

The Top 10 Hottest Days analysis was performed using Apache Pig by sorting the maximum temperature records in descending order and selecting the top ten values. This analysis identifies the ten hottest recorded days in the dataset.

Output of Top 10 Hottest Days:



```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop\pig>hdfs
dfs -cat /weather/output/Top10HottestDays/part-r-00000
USR0000CCLV,20200828,65.6
USR00000ONI,20200129,64.4
USR0000CCLV,20200904,62.8
USR00000ONI,20200514,62.2
USR00000ONI,20200513,61.7
USR0000CCLV,20200830,60.0
ASN00014627,20200429,57.5
USC00026482,20200805,56.1
USC00042319,20200816,54.4
USC00042319,20200712,53.3
```

The following screenshot shows the output containing the 10 days with the highest recorded maximum temperature

7.10. Top 10 Rainiest Days

The Top 10 Rainiest Days analysis was performed using Apache Pig by sorting the rainfall records in descending order and selecting the ten highest rainfall values. The result identifies the days with the highest recorded rainfall in the dataset.

Output of Top 10 Rainiest days:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop\pig>hdfs
dfs -cat /weather/output/Top10RainiestDays/part-r-00000
ASN00083085,20200710,822.8
ASN00087168,20200823,806.8
ASN00058197,20201213,775.0
USW00063888,20200305,634.5
USW00063888,20200320,589.8
ASN00061366,20201018,568.4
ASN00014277,20200111,562.0
ASN00084145,20200403,552.4
ASN00084145,20200315,545.2
USW00063888,20200515,536.7
```

The screenshot below presents the Pig output showing the 10 days with the highest recorded rainfall.

8. MapReduce Analysis

Hadoop MapReduce was used to process the large weather dataset and extract useful weather information. In this project, three different analyses were implemented using MapReduce: **Station Temperature, Station Rainfall, and Extreme Hot Days.**

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>dir classes
Volume in drive C has no label.
Volume Serial Number is 5A2D-B04A

Directory of C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop\classes

09/07/2026  10:30 AM    <DIR>          .
09/07/2026  10:30 AM    <DIR>          ..
09/07/2026  10:30 AM                1,667 WeatherDriver.class
09/07/2026  10:30 AM                1,707 WeatherExtremeHotDriver.class
09/07/2026  10:30 AM                2,184 WeatherExtremeHotMapper.class
09/07/2026  10:30 AM                1,716 WeatherExtremeHotReducer.class
09/07/2026  10:30 AM                2,258 WeatherMapper.class
09/07/2026  10:30 AM                1,700 WeatherRainfallDriver.class
09/07/2026  10:30 AM                2,271 WeatherRainfallMapper.class
09/07/2026  10:30 AM                1,775 WeatherRainfallReducer.class
09/07/2026  10:30 AM                1,759 WeatherReducer.class
               9 File(s)              17,037 bytes
               2 Dir(s) 36,950,589,440 bytes free
```

8.1. Station Temperature Analysis

The Station Temperature analysis was performed using Hadoop MapReduce to process temperature observations from different weather stations. The MapReduce program extracts the required temperature values and processes them based on the weather station.

This analysis helps to understand the temperature information associated with different stations in the dataset.

Output of Station Temperature:

output/MapReduceStationTemperature

Block ID: 1073742955

Block Pool ID: BP-727263249-192.168.246.1-1786856512627

Generation Stamp: 2131

Size: 427735

Availability:

- Mahir

File contents

AE000041196 38.09280303030305

AEM00041194 36.99851301115242

AEM00041217 37.53585657370515

AEM00041218 38.28461538461541

AFM00040938 23.975471698113207

AFM00040990 27.48678678678676

AG000060390 22.95

AG000060590 29.8

The following screenshot shows the MapReduce output generated for the Station Temperature analysis. The output is stored in the MapReduceStationTemperature directory in HDFS.

8.2. Station Rainfall Analysis

The Station Rainfall analysis was implemented using Hadoop MapReduce to process precipitation/rainfall observations from different weather stations. The MapReduce job extracts the rainfall values and aggregates the required information for each station.

This analysis helps to understand the rainfall distribution across the weather stations included in the dataset.

Output of Station Rainfall:

r/output/MapReduceStationRainfall

Block ID: 1073742965

Block Pool ID: BP-727263249-192.168.246.1-1786856512627

Generation Stamp: 2141

Size: 1226335

Availability:

- Mahir

File contents

AE000041196	0.2693452380952381
AEM00041194	0.423876404494382
AEM00041217	8.924999999999999
AEM00041218	18.21428571428571
AFM00040938	3.3345945945945945
AFM00040948	12.25625
AFM00040990	2.9300000000000006
AG000060390	3.898019801980198

The following screenshot presents the MapReduce output for the Station Rainfall analysis. The generated result is stored in the MapReduceStationRainfall directory in HDFS.

8.3. Extreme Hot Days Analysis

The Extreme Hot Days analysis was performed using Hadoop MapReduce to identify weather observations where the maximum temperature reached the defined extreme-hot condition. The MapReduce job processes the temperature records and filters the observations that satisfy the selected temperature threshold.

This analysis helps to identify periods with unusually high temperatures in the weather dataset.

Output of Extreme Hot Days:

```
C:\Users\BS456\Downloads\OneDrive\Desktop\WeatherHadoop>hdfs dfs -cat /weather/output
/MapReduceExtremeHotDays/part-r-00000
AE000041196      190
AEM00041194      176
AEM00041217      166
AEM00041218      171
AFM00040938       15
AFM00040990     106
AG000060590        1
AG000060611        1
AGE00147708        1
AGE00147718        1
AGE00147719       59
AGM00060415        2
AGM00060417        5
AGM00060425        1
AGM00060430        1
AGM00060468        1
AGM00060476        6
AGM00060507        1
AGM00060511        1
AGM00060514        1
AGM00060515        1
AGM00060520        1
```

The following screenshot shows the MapReduce output containing the identified Extreme Hot Days. The result is stored in the MapReduceExtremeHotDays directory in HDFS.

9. MapReduce and Pig Processing

Both MapReduce and Pig were used to perform similar types of weather analysis.

MapReduce provides detailed control over the Map and Reduce stages, while Pig provides a higher-level data-processing approach.

For example, calculating total rainfall using MapReduce requires a Mapper and Reducer to process and aggregate the values. In Pig, the same type of operation can be expressed through a relatively simple sequence of LOAD, FILTER, and aggregation operations.

10. Results And Discussion

The project successfully demonstrated the use of Hadoop for processing and analyzing a large-scale weather dataset. Both **MapReduce** and **Apache Pig** were used to extract useful information from the weather records.

MapReduce Results

Three weather analyses were implemented using MapReduce:

- **Station Temperature:** Processed temperature observations based on weather stations.
- **Station Rainfall:** Processed rainfall/precipitation observations for different stations.
- **Extreme Hot Days:** Identified weather observations that met the defined extreme-temperature condition.

The generated outputs were stored in HDFS and verified after successful execution.

Pig Results

Apache Pig was used to perform ten different weather analyses:

1. Average Maximum Temperature
2. Average Minimum Temperature
3. Extremely Hot Days
4. Hottest Days
5. Monthly Rainfall
6. Monthly Temperature
7. Monthly Extreme Hot Days
8. Top 10 Coldest Days
9. Top 10 Hottest Days
10. Top 10 Rainiest Days

These analyses provided different views of the temperature and rainfall patterns in the dataset. The results demonstrate how Pig can simplify large-scale data processing using high-level data-flow operations.

HDFS output directories generated by MapReduce and Apache Pig analyses:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 17:47	0	0 B	ExtremeHotDays
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 07 10:33	0	0 B	MapReduceExtremeHotDays
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 07 10:23	0	0 B	MapReduceStationRainfall
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 04 11:17	0	0 B	MapReduceStationTemperature
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 18:06	0	0 B	MonthlyExtremeHot
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 16:27	0	0 B	Top10ColdestDays
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 17:23	0	0 B	Top10HottestDays
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 17:07	0	0 B	Top10RainiestDays
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 01 19:59	0	0 B	avg_max_temperature
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 01 20:09	0	0 B	avg_min_temperature
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 16:07	0	0 B	hottest_days
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 02 15:46	0	0 B	monthly_rainfall
☐	drwxr-xr-x	BS456	supergroup	0 B	Sep 01 21:23	0	0 B	monthly_temperature

The following screenshot shows the output directories generated after executing the MapReduce and Apache Pig weather analyses. The results are stored in HDFS under separate directories for the different analyses.

11. Project Challenges

During the implementation of the project, several challenges were encountered.

11.1. Large Dataset

The dataset contains more than 36 million records. Handling such a large dataset requires careful data storage and processing.

11.2. Hadoop Configuration

Hadoop requires proper configuration of services such as HDFS and MapReduce before jobs can be executed successfully.

11.3. HDFS File Management

The dataset needed to be uploaded to the correct HDFS directory before processing. Incorrect paths or missing files can cause Hadoop jobs to fail.

11.4. Pig Execution

Pig scripts must be executed from the correct project directory, and the input/output paths must be correctly specified.

11.5. MapReduce Job Execution

MapReduce jobs can take some time because Hadoop performs different processing stages, including Map, Shuffle, Sort, and Reduce.

These challenges provided practical experience in managing and processing large-scale data using Hadoop.

12. Advantages

The Hadoop-based weather analysis system provides several advantages:

- **Efficient Big Data Processing:** Hadoop can process millions of weather records efficiently.
- **Distributed Processing:** MapReduce distributes data processing into different tasks, making large-scale analysis easier.
- **Scalability:** The system can handle larger datasets when the amount of weather data increases.
- **Reliable Storage:** HDFS provides a suitable storage system for large weather datasets.
- **Multiple Analysis Methods:** The project uses both MapReduce and Apache Pig for weather data analysis.

13. Limitation

The project has some limitations:

- The dataset used in this project is limited to the **2020** time period.
- The analysis mainly focuses on **temperature and rainfall-related information**.
- The project performs data analysis but does not predict future weather conditions.
- The system was implemented in a Hadoop environment rather than as a real-time weather monitoring system.
- Processing such a large dataset can require considerable execution time and system resources.

14. Future Work

The project can be improved in the future by:

- Using **multiple years of weather data** to analyze long-term weather patterns.
- Adding more weather variables such as **wind, snowfall, and other available observations**.
- Applying **machine learning techniques** for weather prediction.
- Extending the system to identify more **extreme weather conditions**.
- Using visualization techniques to present the analyzed weather patterns more clearly.

15. Conclusion

This project demonstrated how **Hadoop** can be used to process and analyze a large-scale weather dataset. HDFS was used for data storage, while **MapReduce and Apache Pig** were used to perform different weather analyses.

The project successfully analyzed temperature and rainfall patterns, including station-based temperature and rainfall information, extreme hot days, monthly weather patterns, and the top hottest, coldest, and rainiest days.

Overall, the project provided practical experience in **Big Data storage, Hadoop processing, MapReduce, Apache Pig, and large-scale weather data analysis**.

16. References

- NOAA National Centers for Environmental Information (NCEI), **Global Historical Climatology Network Daily (GHCN-Daily)**.
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